UNIVERSAL CLOSEDNESS AND PROPERNESS ARE PRESERVED BY THE FACTORIZATION THROUGH A SEPARATED BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays the commutative triangle as part of the statement and refers to it by X, Y and B; the factorization used in the proof is written out.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The graph (or semi-diagonal) is a closed immersion when the target is separated. Lemma lemma-semi-diagonal, used at proof lines 8193-8196. Makes the first factor of X -> Y a closed immersion, hence universally closed.
- Universal closedness is stable under base change. Lemma lemma-base-change-universally-closed, used at proof lines 8197-8200. Makes the projection X x_B Y -> Y universally closed.
- A composite of universally closed morphisms is universally closed. Lemma lemma-composition-universally-closed, used at proof lines 8201-8203. Concludes part (1).
- Permanence statements for separatedness, quasi-compactness and finite typeness. Lemmas lemma-compose-after-separated, lemma-quasi-compact-permanence and lemma-permanence-finite-type, used at proof lines 8204-8207. Upgrade the conclusion of (1) to properness in part (2).

Source reference. Original lemma 03IS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4614–4614. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4687–4687. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03IU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4616–4616. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

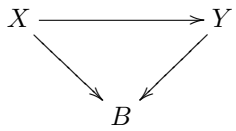
Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0462: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5457–5457. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4684–4684. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Consider a commutative diagram of algebraic spaces*



over S.

(1) *If $X \to B$ is universally closed and $Y \to B$ is separated, then the morphism $X \to Y$ is universally closed. In particular, the image of $|X|$ in $|Y|$ is closed.*
(2) *If $X \to B$ is proper and $Y \to B$ is separated, then the morphism $X \to Y$ is proper.*

Proof. Assume $X \to B$ is universally closed and $Y \to B$ is separated. We factor the morphism as $X \to X \times_B Y \to Y$. The first morphism is a closed immersion, see Lemma 03KO hence universally closed. The projection $X \times_B Y \to Y$ is the base change of a universally closed morphism and hence universally closed, see Lemma 03IS. Thus $X \to Y$ is universally closed as the composition of universally closed morphisms, see Lemma 03IU. This proves (1). To deduce (2) combine (1) with Lemmas 03KR, 03KS, and 0462. □